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(54) **GRAVITY TYPE WATER-PURIFYING DEVICE**

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(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

419,023 A * 1/1890 Roberts B01D 24/183

210/313

445,223 A * 1/1891 Knight B01D 37/025

55/501

(Continued)

FOREIGN PATENT DOCUMENTS

KR 200295215 Y1 11/2002

KR 20040042180 A 5/2004

(Continued)

OTHER PUBLICATIONS

International Search Report issued in PCT/KR2021/019272 dated Mar. 24, 2022, 5 pgs.

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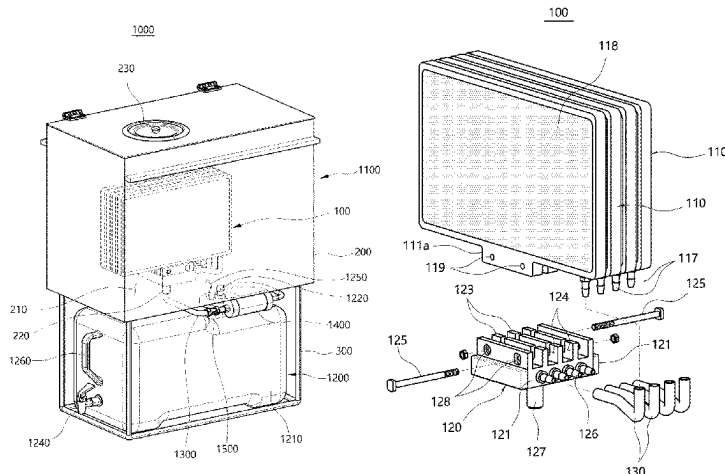
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ABSTRACT

A gravity type water-purifying device is provided. A gravity type water-purifying device according to an embodiment of the present invention comprises: a filtered water generating part comprising a filler module for filtering out foreign materials contained in raw water by using gravity, a housing which stores a predetermined amount of the raw water and has a filtering space formed therein in which the filter module is disposed, and an outlet formed through the housing to allow the filtered water generated through the filter module to be discharged to the outside; and a filtered water storing part which has a filtered water storing space formed therein in order to store a predetermined amount of filtered water generated by the filtered water generating part, is detachably connected to the filtered water generating part by unit of a connection pipe so as to be separable from the

(Continued)



filtered water generating part and be movable by itself, and is provided with an opening/closing unit for discharging filtered water stored in the filtered water storing space to the outside.

13 Claims, 13 Drawing Sheets

(56)

References Cited

U.S. PATENT DOCUMENTS

513,802 A *	1/1894	Luscombe		C02F 1/001	210/476
541,970 A *	7/1895	Harder		C02F 1/001	210/314
546,335 A *	9/1895	Knight		B01D 27/02	210/458
548,487 A *	10/1895	Way		A61M 1/3633	210/411
569,382 A *	10/1896	Knight		B01D 39/1623	210/123
589,887 A *	9/1897	Ziegler		F25D 3/04	210/486
630,846 A *	8/1899	Bliss		B01D 29/33	210/462
645,500 A *	3/1900	Ziegler		B01D 25/26	210/473
676,755 A *	6/1901	Melcher		C02F 1/003	210/473
963,324 A *	7/1910	Randall		B01D 21/26	210/489
1,276,665 A *	8/1918	Leopold		C02F 1/001	210/292
1,401,199 A *	12/1921	Simpson		B01J 20/00	210/505
1,531,095 A *	3/1925	Hoehn		B01D 29/39	210/346
1,674,203 A *	6/1928	Holz		C02F 1/001	210/489
1,723,564 A *	8/1929	Lewis		C02F 1/003	210/287
1,726,035 A *	8/1929	Bomonti		B01D 29/41	210/486
1,919,448 A *	7/1933	Norquist		B01D 29/66	210/313
1,982,924 A *	12/1934	Norquist		B01D 29/39	210/486
2,002,826 A *	5/1935	Moran		B01D 29/39	210/346
2,028,466 A *	1/1936	Moran		B01D 29/39	210/486
2,041,495 A *	5/1936	Schwiers		B01D 29/66	210/420
2,190,305 A *	2/1940	Auberschek		B01D 35/04	210/441
2,306,986 A *	12/1942	Tolman		B01D 39/2003	210/486
2,426,618 A *	9/1947	Klein		B01D 25/26	210/486
2,443,087 A *	6/1948	Ulrich		B01D 29/39	210/486
2,468,296 A *	4/1949	Jacobowitz		B01D 29/39	210/486
2,788,901 A *	4/1957	Boeddinghaus		B01D 29/111	210/486
2,909,285 A *	10/1959	Besler		B01D 29/94	210/330
3,295,687 A *	1/1967	Schmerler		B01D 29/39	55/379
3,339,742 A *	9/1967	Kracklauer		B01D 29/39	210/232
3,339,743 A *	9/1967	Bealle		B01J 47/012	210/256
3,392,837 A *	7/1968	Sanzenbacher		C02F 1/003	210/282
3,430,769 A *	3/1969	Sanzenbacher		C02F 1/003	210/282
3,623,614 A *	11/1971	Schmidt, Jr.		B01D 29/111	210/486
4,968,423 A *	11/1990	McKale		B01D 29/43	210/486
5,035,799 A *	7/1991	Rosberg		B01D 29/66	210/411
5,128,036 A *	7/1992	Svensson		C02F 1/003	210/264
5,476,526 A *	12/1995	Attermeyer		B01D 46/0002	55/496
5,562,824 A *	10/1996	Magnusson		C02F 1/003	210/290
5,928,506 A *	7/1999	Bae		C02F 1/003	210/123
6,013,180 A *	1/2000	Wang		C02F 1/003	210/255
6,129,841 A *	10/2000	Dann		C02F 1/003	210/260
6,248,244 B1 *	6/2001	Dann		C02F 1/003	210/764
D446,791 S *	8/2001	Beckham		D15/1	
6,331,246 B1 *	12/2001	Beckham		B01D 35/30	210/136
6,419,821 B1 *	7/2002	Gadgil		B01D 39/2068	210/256
6,602,425 B2 *	8/2003	Gadgil		B01D 39/2068	210/744
7,014,759 B2 *	3/2006	Radford		C02F 1/003	141/330
7,135,114 B2 *	11/2006	Nonninger		B01D 63/0821	55/482
7,163,618 B2 *	1/2007	Beckham		B01D 35/153	210/473
7,232,517 B1 *	6/2007	Shepherd		C02F 1/003	250/435
7,306,723 B2 *	12/2007	Radford		C02F 1/003	141/330
7,906,019 B2 *	3/2011	Elliott		A01N 59/16	210/205
8,043,502 B2 *	10/2011	Nauta		C02F 1/003	210/473
8,216,465 B2 *	7/2012	Nauta		C02F 1/003	210/473
9,193,605 B2 *	11/2015	Minton-Edison		C02F 1/002	
9,352,252 B2 *	5/2016	Vestergaard Frandsen		B01D 29/62	
9,352,979 B2 *	5/2016	Kuennen		C02F 3/06	
9,975,068 B1 *	5/2018	Maresh		B01D 24/00	
10,336,639 B2 *	7/2019	Kuennen		C02F 1/281	
10,589,195 B2 *	3/2020	Han		C02F 1/004	
10,589,199 B2 *	3/2020	See		C02F 1/444	
10,589,200 B1 *	3/2020	Maresh		C02F 1/001	
10,647,592 B2 *	5/2020	Geelhood		C02F 1/002	
10,905,976 B2 *	2/2021	Johnson		B01D 21/0012	
11,161,062 B2 *	11/2021	Johnson		C02F 9/20	
11,202,975 B2 *	12/2021	Han		C02F 1/004	
11,247,153 B2 *	2/2022	Han		C02F 1/44	
11,872,506 B2 *	1/2024	Weitz		B01D 24/38	
11,938,424 B2 *	3/2024	Han		C02F 1/003	
12,311,292 B2 *	5/2025	Song		B01D 29/15	
12,318,720 B2 *	6/2025	Johnson		B01D 29/58	
2002/0113022 A1 *	8/2002	Gadgil		B01D 39/2068	210/748.11
2003/0213733 A1 *	11/2003	Beckham		A61M 1/79	210/500.1
2004/0134932 A1 *	7/2004	Lobdell		B67D 3/0029	222/189.06
2006/0144781 A1 *	7/2006	Carlson		B01D 61/20	210/473
2007/0080118 A1 *	4/2007	Lindsley		B01D 35/0276	210/791

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0217252 A1* 9/2008 Elliott A01N 59/16
210/661
2011/0303589 A1* 12/2011 Kuennen C02F 1/52
210/120
2012/0187052 A1* 7/2012 Elliott A01N 59/16
210/764
2012/0267314 A1* 10/2012 Minton-Edison C02F 1/44
210/695
2014/0144829 A1* 5/2014 Takeda B01D 63/024
210/295
2014/0216993 A1* 8/2014 Pradeep B01D 29/00
210/123
2015/0284260 A1* 10/2015 Huda C02F 1/003
210/128
2016/0114271 A1* 4/2016 Michaud C02F 1/003
210/807
2016/0236958 A1* 8/2016 Kuennen C02F 1/283

2018/0154290 A1* 6/2018 See C02F 1/002
2019/0209972 A1* 7/2019 Han B01D 63/0821
2019/0247772 A1* 8/2019 Han B01D 69/06
2020/0071189 A1* 3/2020 Geelhood C02F 1/283
2020/0215463 A1* 7/2020 Johnson B01D 35/027
2022/0047975 A1* 2/2022 Johnson B01D 35/02
2022/0054958 A1* 2/2022 Han B01D 36/001
2023/0166203 A1* 6/2023 Song C02F 1/001
210/455
2024/0066450 A1* 2/2024 Lee B01D 29/00

FOREIGN PATENT DOCUMENTS

KR 20090129389 A 12/2009
KR 1020150011745 A 2/2015
KR 101516969 B1 4/2015
KR 20180116667 A 10/2018
KR 20200006135 A 1/2020
KR 20200071418 A 6/2020

* cited by examiner

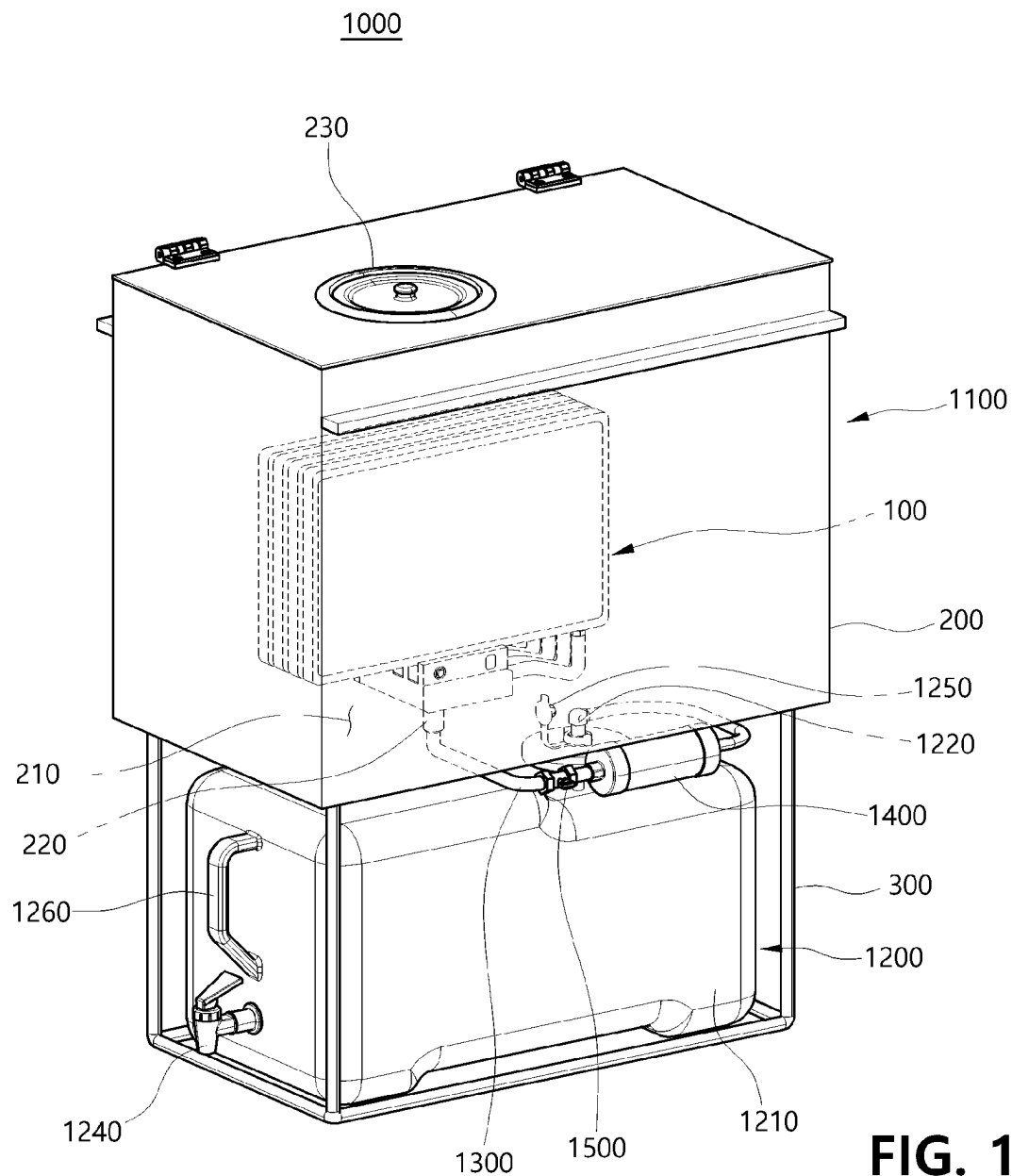


FIG. 1

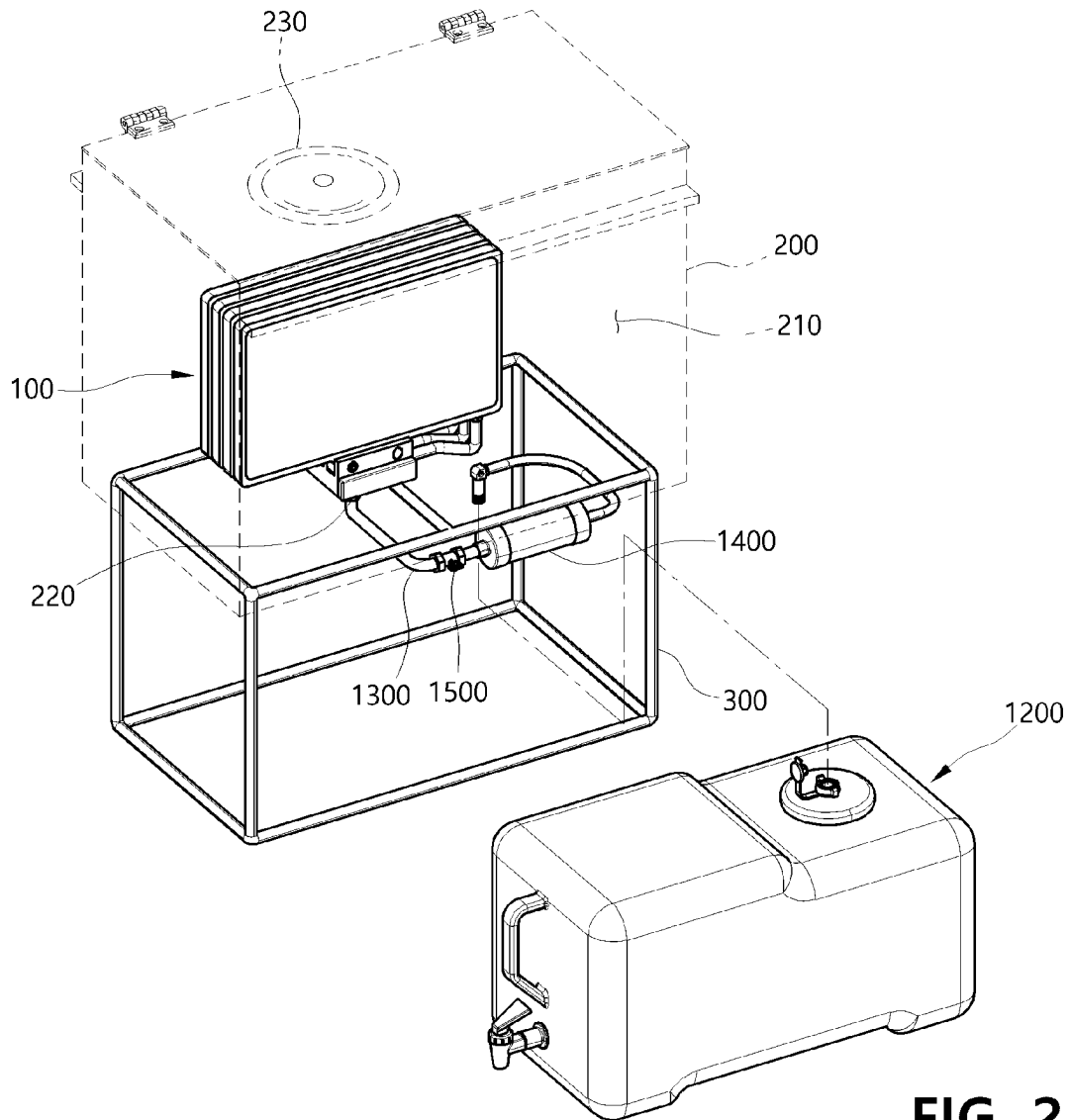


FIG. 2

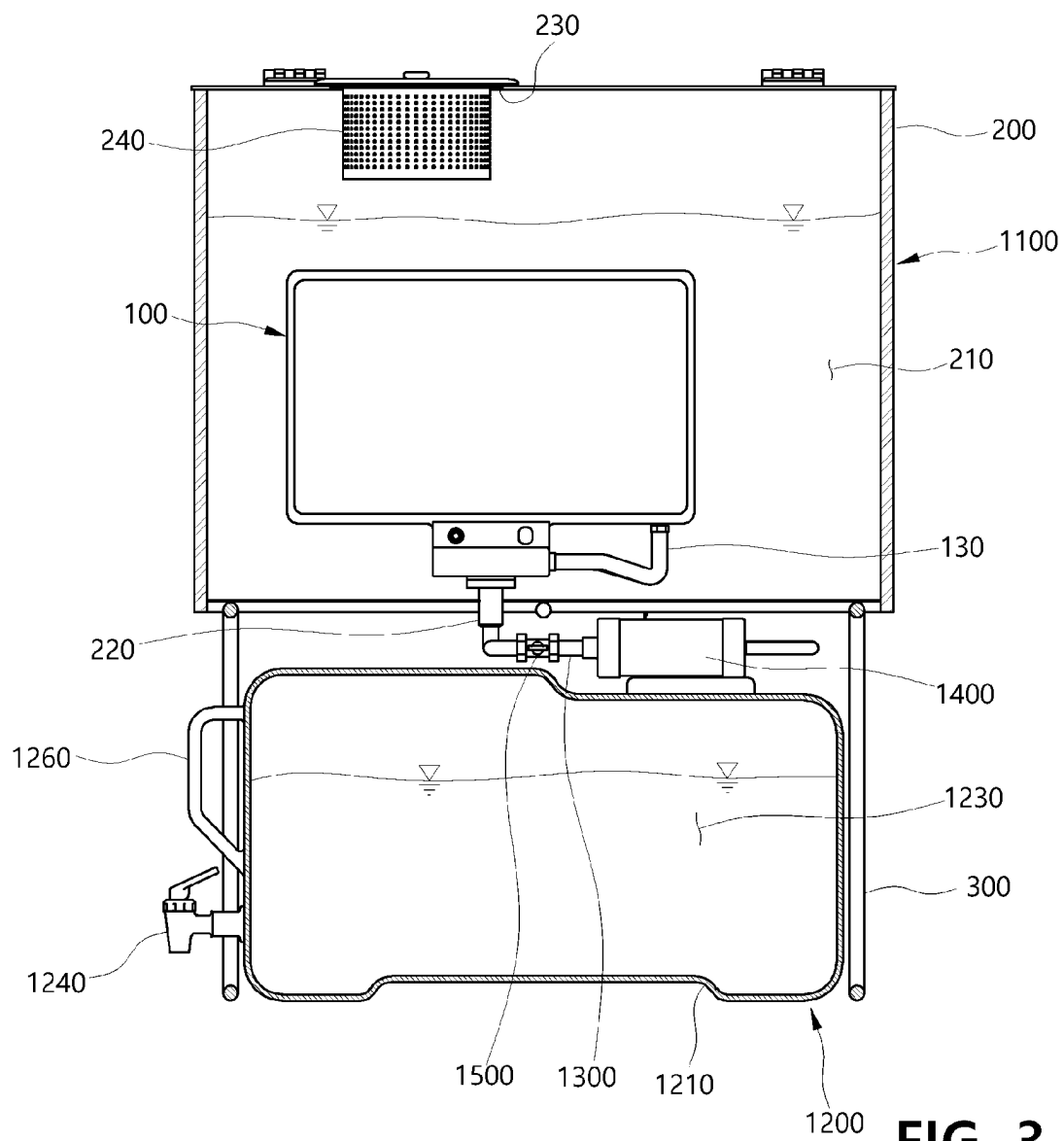


FIG. 3

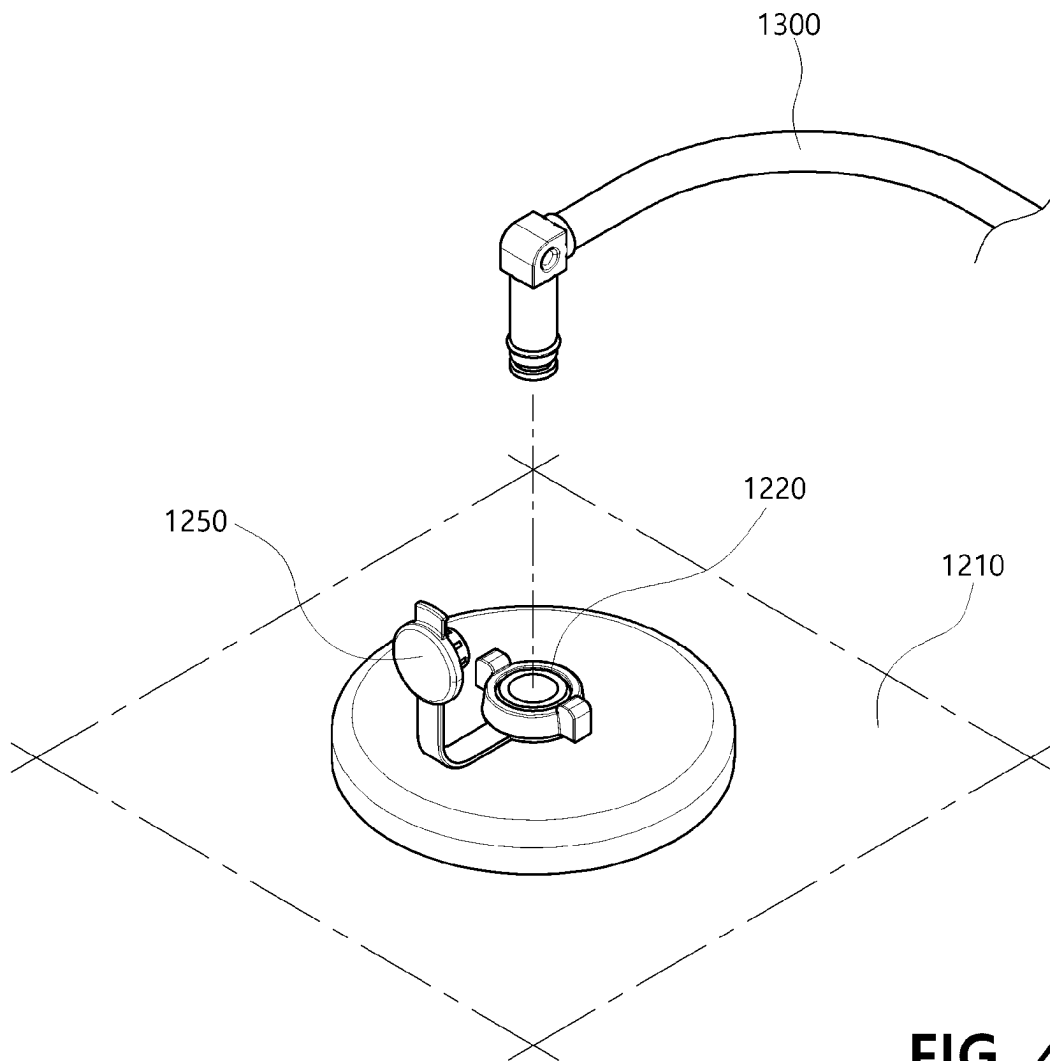
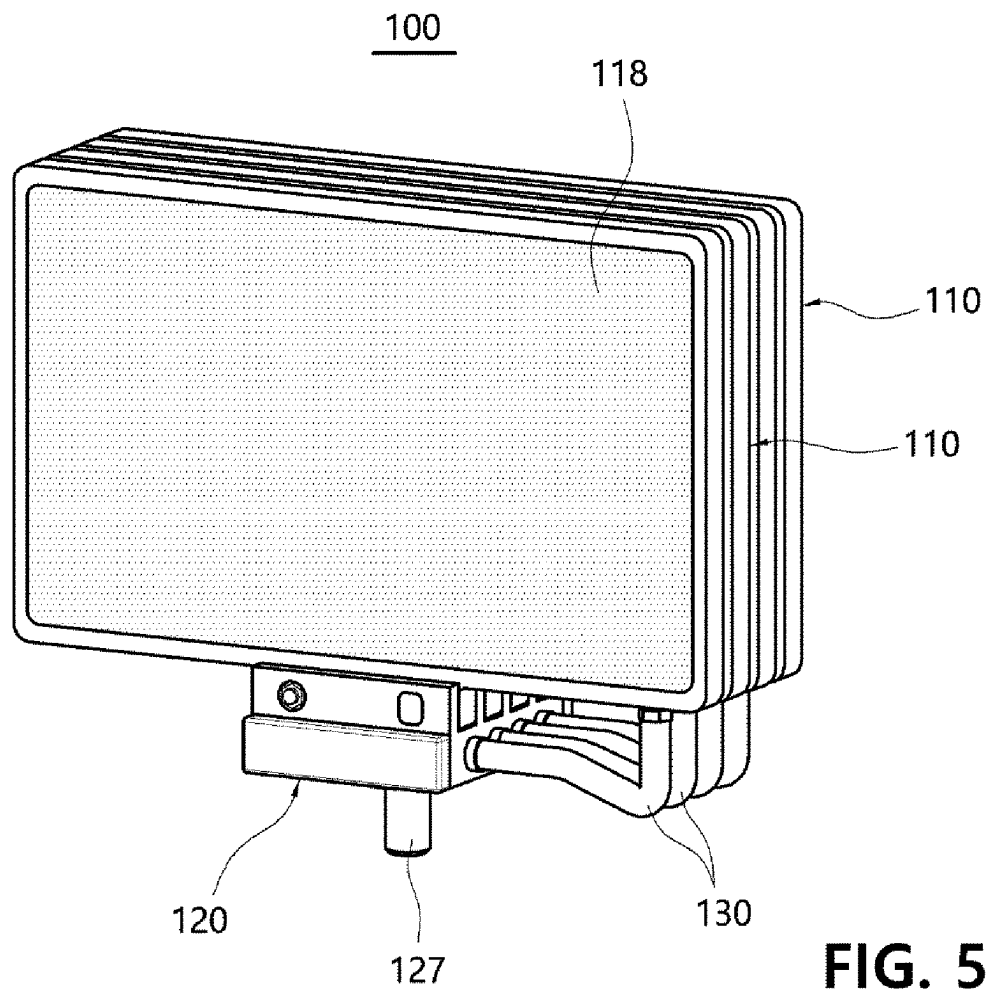


FIG. 4

**FIG. 5**

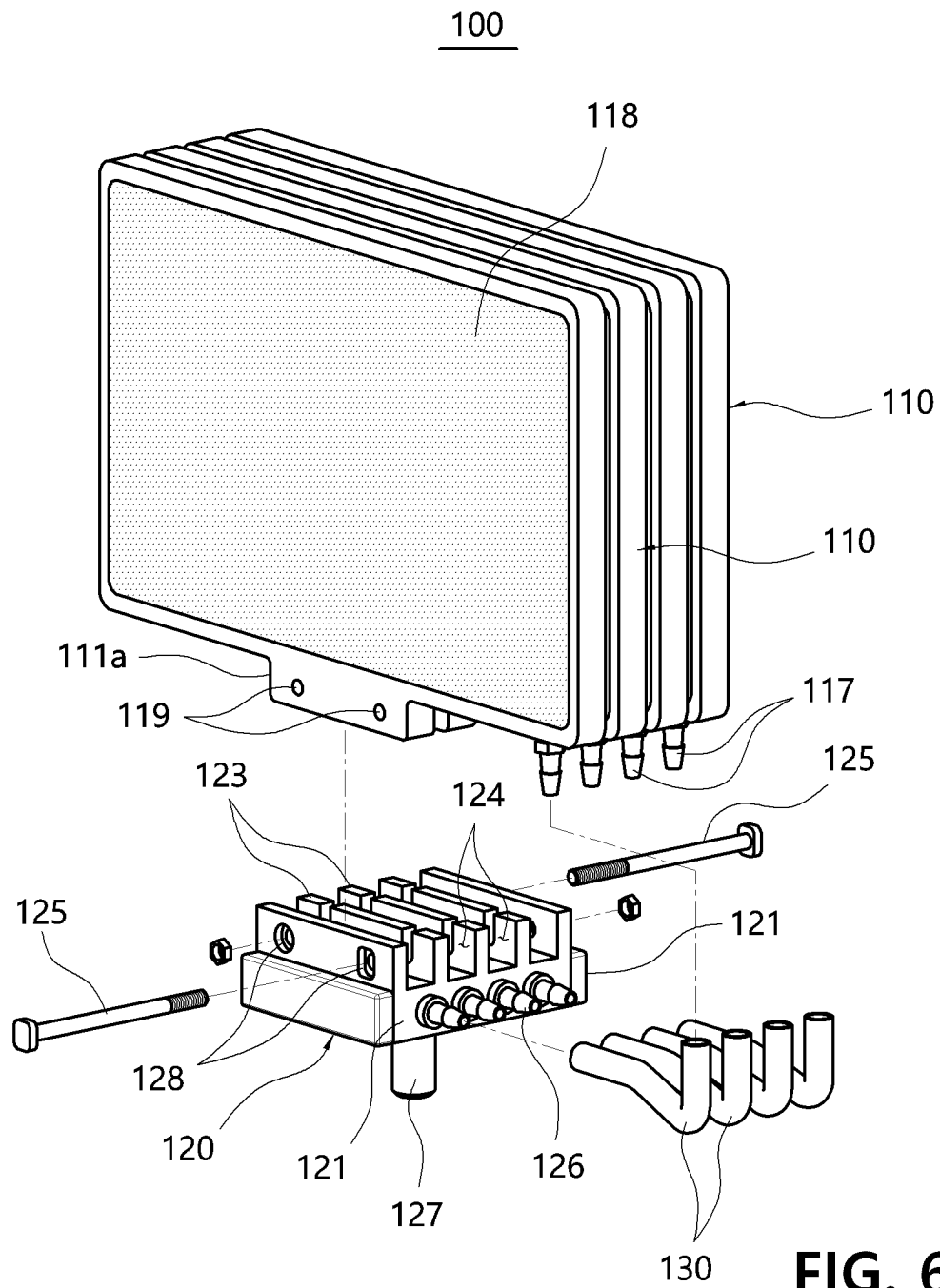
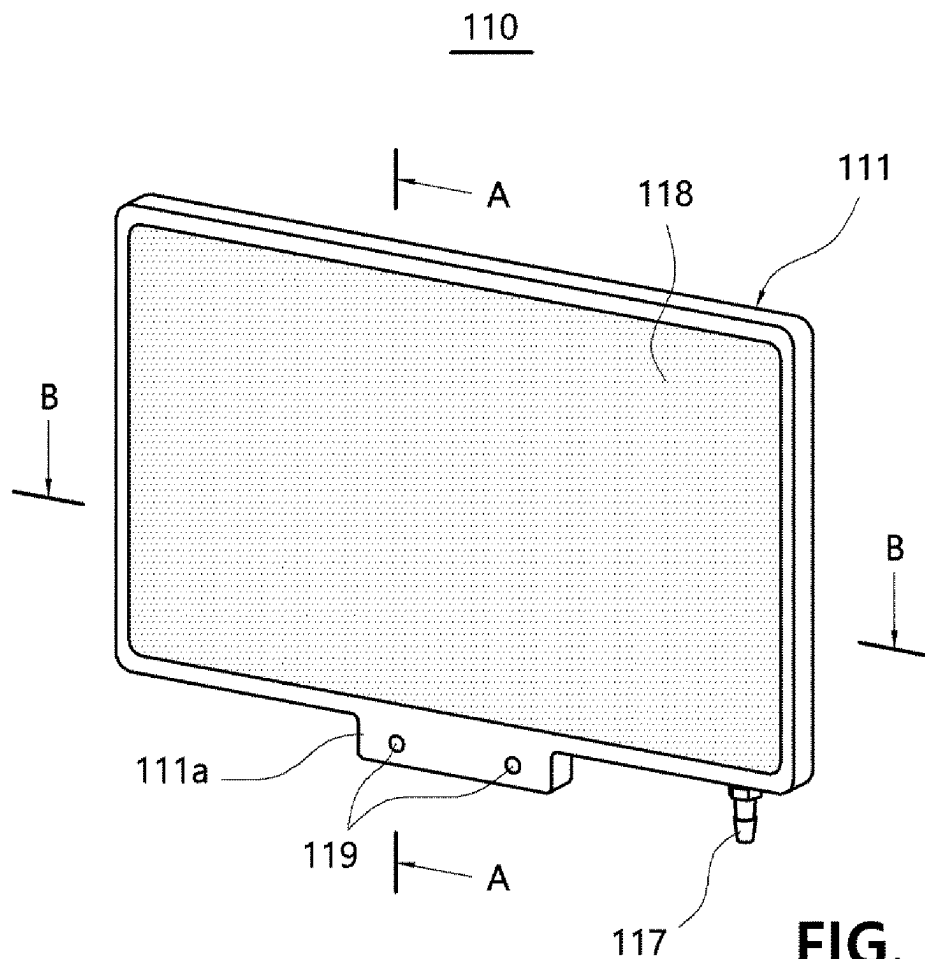


FIG. 6

**FIG. 7**

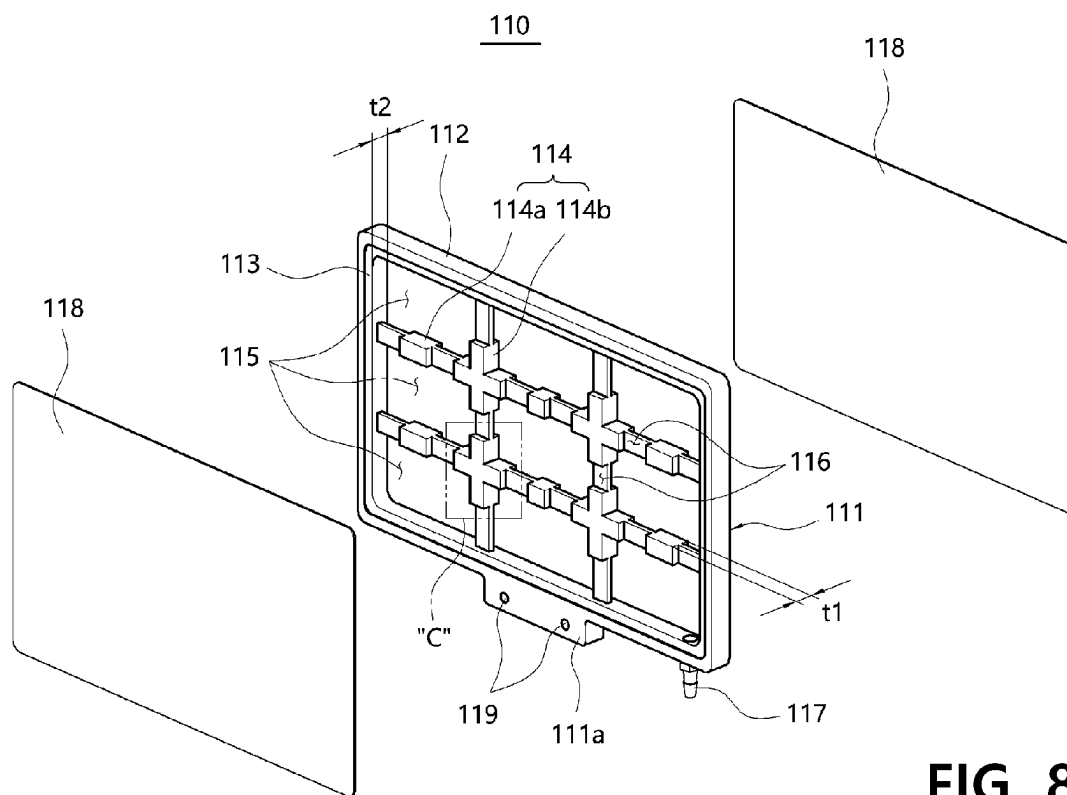


FIG. 8

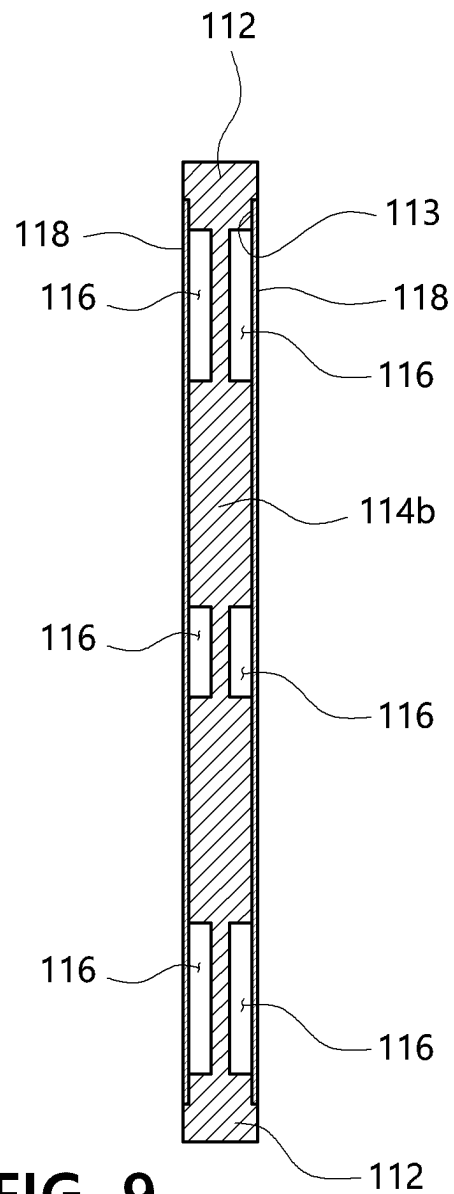


FIG. 9

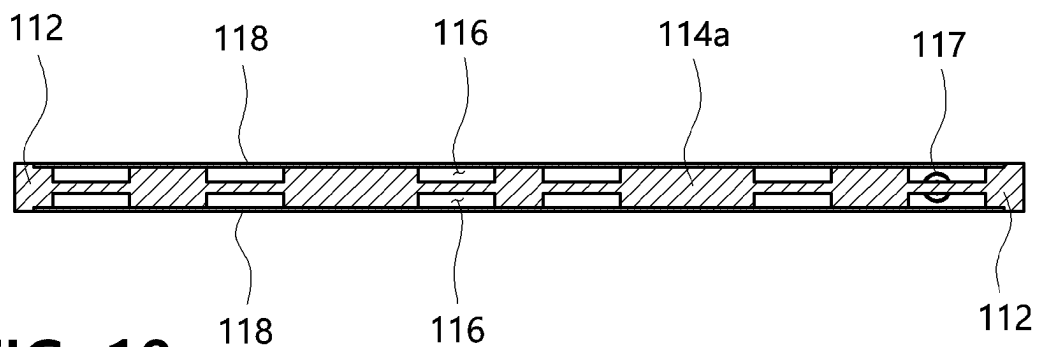


FIG. 10

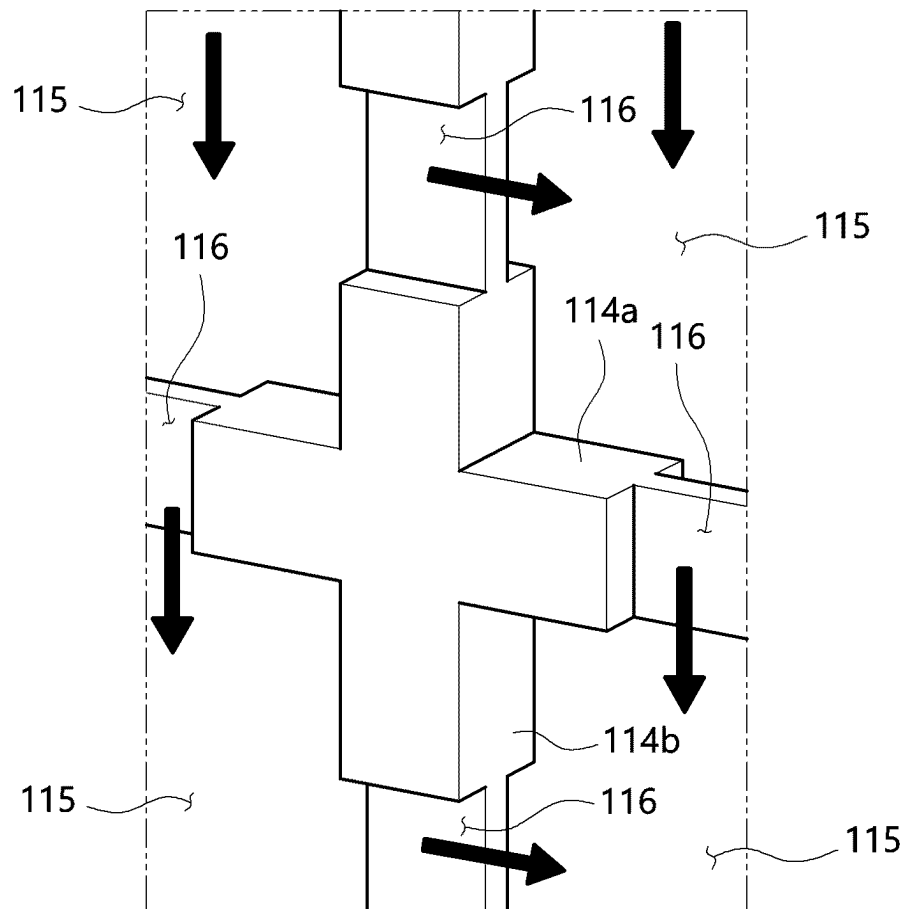
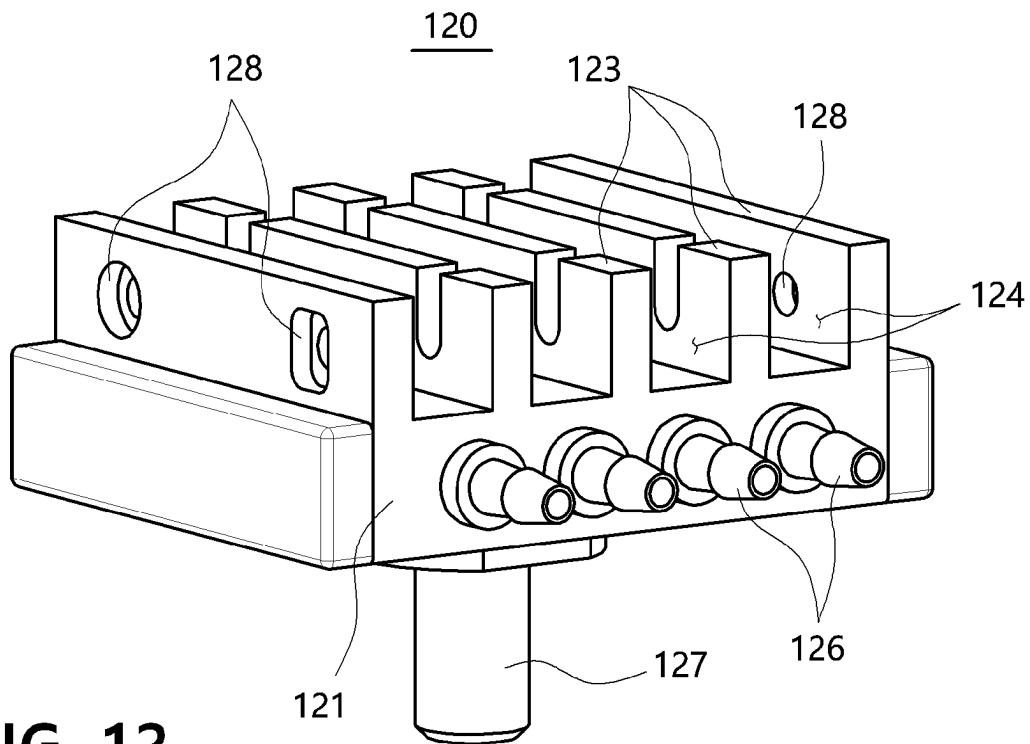
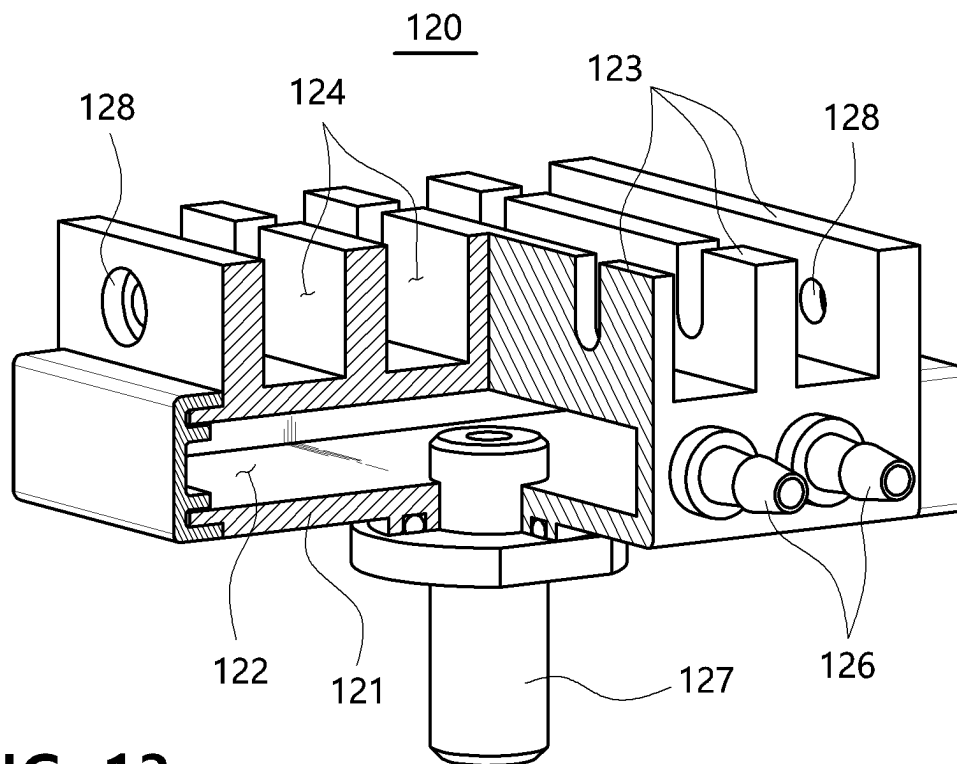


FIG. 11

**FIG. 12**

**FIG. 13**

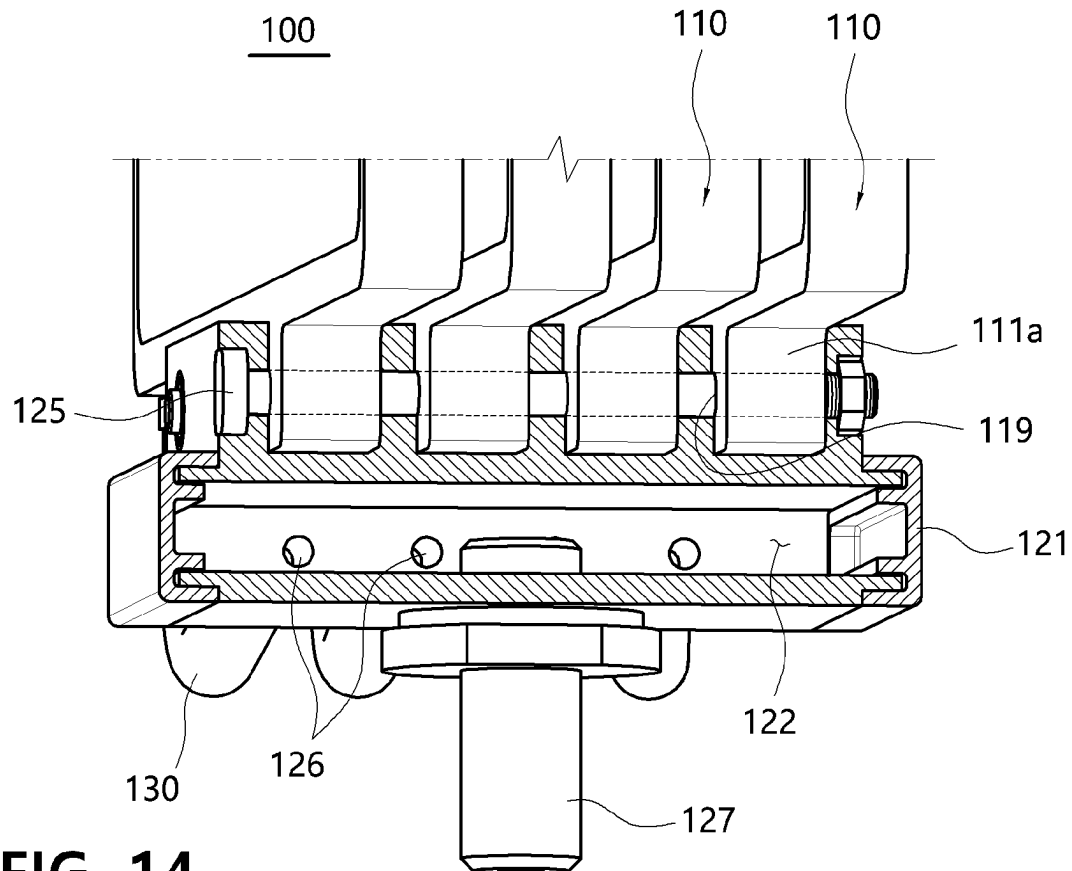


FIG. 14

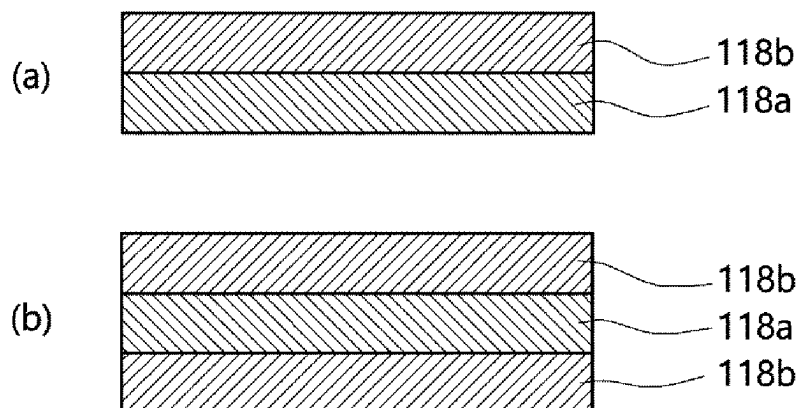


FIG. 15

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**GRAVITY TYPE WATER-PURIFYING
DEVICE****CROSS REFERENCE TO RELATED
APPLICATION**

This application is a 35 U.S.C. 371 National Phase Entry Application from PCT/KR2021/019272 filed Dec. 17, 2021, which claims priority to and the benefit of Korean Patent Application No. 10-2020-0183243, filed on Dec. 24, 2020, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention relates to a gravity-type water-purifying device.

BACKGROUND ART

Supplying clean and safe water is essentially necessary for health and survival of humans. However, since clean water is not sufficiently supplied in Africa or developing countries, people frequently use water in a contaminated pond or puddle as drinking water.

When drinking contaminated water, the drinker may get various infectious diseases due to microbes included in the contaminated water. That is, there is a problem that the microbes contained in the contaminated water cause illness such as abdominal pain, diarrhea, vomiting and the like, and in a severe case threaten the life of the drinker.

Accordingly, in order to prevent such problem, it is important to ensure that the food or water we eat is not contaminated by microbes.

Currently, since the overall mortality rate can be reduced merely if clean water and sufficient food are secured in developing countries, it can be said that the effect of environmental improvement, which can be achieved by supplying clean water, is far greater than advances in medicine or treatment technology.

However, since a conventional water purifier passes raw water through a filter by applying a certain water pressure to filter out foreign substances contained in the raw water, the purifier requires a pump facility capable of imparting the certain water pressure, and electrical facilities for operating the pump facility must also be provided. In the absence of such electrical facilities, there is a limitation that the conventional water purifier cannot be used.

DISCLOSURE**Technical Problem**

The present invention has been devised in view of the above points, and an object of the present invention is to provide a gravity-type water-purifying device capable of conveniently producing filtered water by using gravity.

Another object of the present invention is to provide a gravity-type water-purifying device capable of reducing downtime by configuring a storage tank in which filtered water is stored in a movable manner so that it can be individually separated.

Technical Solution

In order to achieve the above objects, the present invention provides a gravity-type water-purifying device com-

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prising: a filtered water production part including: a filter module for filtering out foreign substances contained in raw water by using gravity, a housing in which a filtration space is formed to store a certain amount of the raw water and to dispose the filter module therein, and an outflux formed in the housing to allow the filtered water produced through the filter module to be discharged to the outside; and a movable filtered water storage part in which a filtered water storage space is formed to store a certain amount of the filtered water produced from the filtered water production part, wherein the filtered water storage part is detachably connected to the filtered water production part through a connection pipe so as to be separable from the filtered water production part and be movable by itself, and is provided with an opening/closing unit for discharging the filtered water stored in the filtered water storage space to the outside.

In addition, the movable filtered water storage part may include a box-shaped storage tank including the filtered water storage space, and a fitting formed in the storage tank so as to be detachably coupled to an end of the connection pipe.

In addition, the connection pipe may further include a filter for re-filtering the filtered water moving from the outflux to the movable filtered water storage part.

In addition, the connection pipe may further include an opening/closing valve for allowing or blocking the movement of the filtered water from the outflux to the movable filtered water storage part.

In addition, the filter module may comprise: a plurality of plate-shaped filter members, which are formed in a plate shape having a predetermined area, produce filtered water from raw water introduced from the outside by gravity, and are spaced apart from each other in parallel along one direction while one surface of each filter member faces each other at a predetermined interval; and a filtered water integration member for fixing one side of each of the plurality of plate-shaped filter members so that the plurality of plate-shaped filter members can be kept spaced apart from each other along one direction while facing each other, and for integrating and discharging the filtered water individually produced from each of the plurality of plate-shaped filter members.

In addition, each of the plurality of plate-shaped filter members may include: a support frame comprising a loop-shaped edge member having an empty space open on both sides, and a partition member coupled to the edge member to partition the empty space into a plurality of storage spaces; a pair of filtration members formed in a plate shape having a predetermined area and coupled to both surfaces of the support frame to produce filtered water from raw water moving from the outside to the plurality of storage spaces; at least one communication path formed in the partition member to allow the plurality of storage spaces to communicate with each other; and a receiving port provided on one side of the support frame so as to communicate with any one of the plurality of storage spaces and connected to the filtered water integration member via a tube.

In addition, the communication path may be an accommodation groove formed to be recessed into the partition member.

In addition, the edge member may include a stepped portion formed along the inner side of the edge, and an edge side of the filtration member may be fixed to the stepped portion.

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In addition, the partition member may include at least one lateral member having a predetermined length and at least one longitudinal member disposed to intersect the lateral member.

In addition, the filtration member may include a plate-shaped support having a predetermined area, and a nanofiber web layer formed of nanofiber and disposed on one surface or both surfaces of the support.

In addition, all of the plurality of plate-shaped filter members may be provided with the same filter member.

In addition, the filtered water integration member may include: a body having an integration space for integrating the filtered water incoming from each of the plurality of plate-shaped filter members; a plurality of protrusions protruding from one surface of the body to form a plurality of accommodation grooves into which one side of the plurality of plate-shaped filter members are respectively inserted; and a fastening member fastened to the body and passing through all of the plurality of protrusions so as to prevent each of the plate-shaped filter members respectively inserted into the plurality of accommodation grooves from being separated from the accommodation grooves.

In addition, the body may further include a plurality of inlets connected one-to-one to the receiving ports through a tube wherein the receiving ports are respectively provided in the plurality of plate-shaped filter members so as to introduce the filtered water individually produced by the plurality of plate-shaped filter members into the integration space, and at least one outlet for discharging the filtered water from the integration space to the outside.

Advantageous Effects

According to the present invention, filtered water can be simply produced by using gravity even under poor conditions in which power such as electricity is not supplied, thereby obtaining clean filtered water even in a poor environment. As such, it is possible to simply obtain filtered water even in a poor environment, thereby preventing diseases caused by ingestion of contaminated water and improving the quality of life.

In addition, according to the present invention, a storage tank in which filtered water is stored is configured in a movable manner so that it can be individually separated, whereby downtime can be reduced to increase the production of the filtered water.

DESCRIPTION OF DRAWINGS

FIG. 1 is a view showing a gravity-type water-purifying device according to an embodiment of the present invention.

FIG. 2 is a view showing a state in which a movable filtered water storage part in FIG. 1 is separated.

FIG. 3 is a front view showing a housing and a movable filtered water storage part in FIG. 1 partially cut away.

FIG. 4 is a view showing a connection relationship between a connection pipe and a storage tank that can be applied to a gravity-type water-purifying device according to an embodiment of the present invention.

FIG. 5 is a view showing a filter module extracted from FIG. 1.

FIG. 6 is an exploded view of FIG. 5.

FIG. 7 is a view showing a plate-shaped filter member applicable to a filter module in a gravity-type water-purifying device according to an embodiment of the present invention.

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FIG. 8 is a view showing a state in which the filtration member in FIG. 7 is separated.

FIG. 9 is a cross-sectional view taken along line A-A of FIG. 7.

FIG. 10 is a cross-sectional view taken along line B-B of FIG. 7.

FIG. 11 is an enlarged view of part C in FIG. 8.

FIG. 12 is a view showing a filtered water integration member applicable to a filter module in a gravity-type water-purifying device according to an embodiment of the present invention.

FIG. 13 is a view showing a state in which a part in FIG. 12 is cut away.

FIG. 14 is a view showing a coupling relationship between a filtered water integration member and a plate-shaped filter member in a gravity-type water-purifying device according to an embodiment of the present invention.

FIG. 15 is a schematic diagram illustrating a detailed configuration of a filtration member applicable to a plate-shaped filter member in a gravity-type water-purifying device according to an embodiment of the present invention.

MODES OF THE INVENTION

Hereinafter, with reference to the accompanying drawings, embodiments of the present invention will be described in detail so as to be easily implemented by one of ordinary skill in the art to which the present invention pertains. The present invention may be embodied in a variety of forms and is not be limited to the embodiments described herein. In order to clearly describe the present invention in the drawing, parts irrelevant to the description are omitted from the drawings; and throughout the specification, same or similar components will be referred to as like reference numerals.

A gravity-type water-purifying device **1000** according to an embodiment of the present invention may produce filtered water from raw water by using water pressure due to gravity or potential energy.

Herein, the filtered water may be used as potable water or drinking water.

In this case, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may include a filter module **100** for producing filtered water from raw water, and the filtered water produced from the filter module **100** may be moved toward a storage tank **1210** through a connection pipe **1300** and be stored in the storage tank **1210**. In addition, the storage tank **1210** may be detachably coupled to the connection pipe **1300**.

Accordingly, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the storage tank **1210** may be easily separated from the connection pipe **1300** to be replaced with another storage tank **1210**.

That is, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, when the storage tank **1210** is full of the filtered water or the storage tank **1210** needs to be washed, the storage tank **1210** connected to the connection pipe **1300** may be easily replaced with another storage tank.

Through this, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, downtime during which the filtered water cannot be produced may be reduced, and contamination of the filtered water that may occur in the process of transferring the filtered water produced from the filter module **100** to another storage container may be prevented in advance.

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To this end, the gravity-type water-purifying device **1000** according to an embodiment of the present invention includes a filtered water production part **1100** and a movable filtered water storage part **1200** as shown in FIGS. 1 to 3, and the filtered water production part **1100** may be connected to the movable filtered water storage part **1200** via the connection pipe **1300**.

The filtered water production part **1100** may produce the filtered water from raw water by using gravity as described above.

To this end, the filtered water production part **1100** may include a filter module **100** and a housing **200**.

Specifically, the housing **200** may include a filtration space **210** in which raw water to be treated is stored, and the filter module **100** may be disposed in the filtration space **210**.

For example, as shown in FIG. 3, the housing **200** may be formed in a box shape having an internal space of a predetermined volume, and the internal space may be the filtration space **210** in which a certain amount of the raw water is stored and the filter module **100** is disposed.

Accordingly, when the filtration space **210** is filled with the raw water, at least a portion of the filter module **100** may be submerged in the raw water filled in the filtration space **210** as shown in FIG. 3.

In this case, the filtered water production part **1100** may include an outflux **220** for discharging the filtered water produced from the filter module **100** to the movable filtered water storage part **1200**.

For example, the outflux **220** may be formed on the bottom surface of the housing **200**, and the outlet **127** of the filter module **100** may be coupled to the outflux **220**. In addition, the end of the connection pipe **1300** interconnecting the filtered water production part **1100** and the movable filtered water storage part **1200** may be coupled to the outflux **220**.

In this case, the outflux **220** may serve both as a role of discharging the filtered water produced from the filter module **100** to the outside and as a fastener for fastening the outlet **127** of the filter module **100**.

Accordingly, when the filtered water produced from the filter module **100** is discharged from the filtration space **210** through the outflux **220**, the filtered water may move to the movable filtered water storage part **1200** along the connection pipe **1300**. Accordingly, the water level of the raw water filled in the filtration space **210** may be lowered, and the pressure due to gravity generated as the water level of the raw water is lowered may be applied to the filter module **100**.

As a result, the filter module **100** may produce filtered water by filtering out foreign substances from the raw water by using the gravity generated when the water level of the raw water changes. A detailed description of the filter module **100** will be provided later.

As such, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may produce filtered water by using the filter module **100** in the filtration space **210**, and the filtered water produced in the filtration space **210** by the filter module **100** may move to the movable filtered water storage part **1200** through the outflux **220**.

As a result, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may easily produce filtered water without using additional power such as electricity, whereby filtered water may be easily produced even in a harsh environment where infrastructure such as electricity is not installed or electricity is difficult to supply.

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In the drawings and description, it has been shown and described that the outflux **220** for discharging the filtered water produced from the filter module **100** to the movable filtered water storage part **1200** is formed on the bottom surface of the housing **200**, but the present invention is not limited thereto, and the position of the outflux **220** may be changed to an appropriate location according to design conditions in consideration of the connection relationship with the filter module **100**.

For example, if the outflux **220** may move the filtered water produced from the filter module **100** to the movable filtered water storage part **1200** through the connection pipe **1300** without flowing out to the filtration space **210**, it may be formed in the housing **200** so as to be located in an appropriate position.

In this case, the housing **200** may further include an inlet **230** for supplying the raw water to the filtration space **210**. In this case, as shown in FIG. 3, a filtering net **240** may be detachably coupled to the inlet **230**. Such a filtering net **240** may filter out bulky foreign substances such as fallen leaves or tree branches included in raw water when the raw water is supplied to the filtration space **210**.

Meanwhile, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may further include a support part **300** for supporting the housing **200**.

Such a support part **300** may support one side of the housing **200** so that the housing **200** may be disposed in a state of being spaced apart from the bottom surface by a certain height.

For example, the support part **300** may be a frame structure formed to have a predetermined height, and may support the lower surface of the housing **200**.

Herein, the support part **300** may be configured separately from the housing **200**, or may be formed integrally with the housing **200**.

As described above, the movable filtered water storage part **1200** may be connected to the filtered water production part **1100** via a connection pipe **1300** having one end connected to the outflux **220**.

Accordingly, the movable filtered water storage part **1200** may store a certain amount of the filtered water produced from the filter module **100** therein.

To this end, the movable filtered water storage part **1200** may include a storage tank **1210** having a filtered water storage space **1230** for storing a certain amount of the filtered water introduced from the filtered water production part **1100**, and the storage tank **1210** may include a fitting **1220** to which an end of the connection pipe **1300** is connected.

Accordingly, the filtered water produced from the filtered water production part **1100** may be introduced into the filtered water storage space **1230** through the connection pipe **1300**.

In this case, as shown in FIG. 4, an end of the connection pipe **1300** may be detachably coupled to the fitting **1220**.

Accordingly, when the storage tank **1210** needs to be washed or a large amount of the filtered water is filled in the filtered water storage space **1230**, the user may easily replace the storage tank **1210** with another storage tank by separating the end of the connection pipe **1300** from the fitting **1220**.

As a result, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the storage tank **1210** may be easily replaced through the coupling and separation of the connection pipe **1300** and the

fitting **1220**, whereby the filtered water may be continuously produced through the filter module **100**.

That is, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the storage tank **1210** may be individually separated and combined with the connection pipe **1300**, thereby reducing downtime during which the production of filtered water is stopped.

In this case, as shown in FIG. 4, the storage tank **1210** may further include a stopper **1250** for opening and closing the fitting **1220**, and a handle **1260** for allowing the user to easily handle the storage tank **1210**.

In addition, the storage tank **1210** may further include an opening/closing unit **1240** for discharging the filtered water filled in the filtered water storage space **1230** to the outside through a user's manipulation. For example, the opening/closing unit **1240** may be a well-known opening/closing valve.

Accordingly, when a certain amount of filtered water is filled in the filtered water storage space **1230**, the user may separate the connection pipe **1300** from the fitting **1220** and then cover the fitting **1220** with the stopper **1250**.

Through this, the stopper **1250** may prevent foreign substances from flowing into the filtered water storage space **1230** from the outside.

In addition, the user may conveniently move the storage tank **1210** to a desired position by using the handle **1260** in a state in which the filtered water storage space **1230** is filled with a certain amount of filtered water.

Further, when the user wants to use the filtered water, the filtered water stored in the filtered water storage space **1230** may be taken out by manipulating the opening/closing unit **1240**. Thus, the user may conveniently use the filtered water stored in the filtered water storage space **1230**.

As described above, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the movable filtered water storage part **1200** has the storage tank **1210** implemented as a portable water bottle type so that it may be moved and stored individually, thereby improving user convenience.

In addition, when the user wants to use the filtered water produced from the filter module **100** in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the user may directly take out the filtered water stored in the filtered water storage space **1230** through the opening/closing unit **1240** provided in the storage tank **1210**.

Accordingly, since it is not necessary to move the filtered water filled in the storage tank **1210** to a separate storage container for use of the filtered water, contamination of the filtered water that may occur in the process of moving the filtered water to a separate storage container may be prevented in advance.

Meanwhile, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may further include an opening/closing valve **1500** in the connection pipe **1300** for allowing or blocking the movement of filtered water produced from the filter module **100** toward the movable filtered water storage part **1200**.

That is, when the storage tank **1210** needs to be replaced, the user may block the movement of filtered water produced from the filter module **100** toward the movable filtered water storage part **1200** along the connection pipe **1300** by manipulating the opening/closing valve **1500**.

In such a case, since the discharge of the filtered water through the outflux **220** from the filtration space **210** is blocked, the production of filtered water using the filter module **100** may be stopped.

Accordingly, the user may temporarily stop the production of filtered water by manipulating the opening/closing valve **1500** if necessary, whereby even if the end of the connection pipe **1300** is separated from the fitting **1220** of the storage tank **1210**, it may be minimized that the filtered water is discarded to the outside from the end of the connection pipe **1300** separated from the fitting **1220**.

In this case, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may further include a filter **1400** to further improve the quality of the filtered water by filtering the filtered water produced by the filtered water production part **1100** again.

Such a filter **1400** may secondarily filter the filtered water produced by the filter module **100**. Accordingly, the filtered water may be secondarily filtered through the filter **1400** and then moved to the filtered water storage space **1230** of the storage tank **1210**.

To this end, the filter **1400** may be provided in the connection pipe **1300** connecting the filtered water production part **1100** and the storage tank **1210** to each other.

For example, the filter **1400** may improve the quality of the filtered water by removing the odor of the filtered water produced through the filter module **100** and/or harmful chemicals included in the filtered water.

As a non-limiting example, the filter **1400** may be a known carbon filter, but the type of the filter **1400** is not limited thereto, and any of the known various filters may be used without limitation as long as it may improve the quality of the filtered water produced from the filter module **100** so that the filtered water may be used as potable water or drinking water. In addition, the filter **1400** may be changed to a suitable type depending on a place where the gravity-type water-purifying device **1000** according to an embodiment of the present invention is installed or used.

Accordingly, the filtered water produced through the filter module **100** in the filtered water production part **1100** may be filtered again while passing through the filter **1400** in the process of moving to the storage tank **1210** side along the connection pipe **1300**.

As a result, the filtered water in a state of being filtered twice through the filter module **100** disposed in the filtration space **210** and the filter **1400** provided in the connection pipe **1300** may be stored in the storage tank **1210**.

Meanwhile, as described above, the filter module **100** may produce the filtered water by filtering out foreign substances included in raw water using gravity.

As such a filter module **100**, any known various filter modules may be used without limitation as long as they may filter out foreign substances from raw water, but preferably may be configured to increase the production of filtered water.

As a non-limiting example, the filter module **100** may include a plurality of plate-shaped filter members **110** and a filtered water integration member or manifold **120** as shown in FIGS. 5 and 6.

The plurality of plate-shaped filter members **110** may produce the filtered water from the raw water by using water pressure caused by gravity or potential energy in a state in which at least a portion thereof is submerged in the raw water.

In this case, each of the plurality of plate-shaped filter members **110** may be formed in a plate shape having a predetermined area. In addition, the plurality of plate-shaped

filter members **110** may be disposed so that one surface of each filter member faces each other at a predetermined interval, and may be spaced apart from each other in parallel along one direction.

Accordingly, since each of the plate-shaped filter members **110** in the filter module **100** may individually produce filtered water, the production of filtered water per unit time may be increased.

In this case, each of the plurality of plate-shaped filter members **110** may be provided in a different form, but may be provided in the same form.

Through this, all of the plurality of plate-shaped filter members **110** in the filter module **100** may be manufactured by using one mold or manufacturing facility, thereby reducing the production cost for producing the plate-shaped filter members.

In addition, the total number of use of the plate-shaped filter member **110** in the filter module **100** may be easily adjusted according to the amount of filtered water required per hour, thereby enhancing the convenience of use.

The plate-shaped filter member **110** may include a support frame **111**, a pair of filtration members **118**, a communication path **116** and a receiving port **117** as shown in FIGS. 7 to 10.

The support frame **111** may support the pair of filtration members **118** and also provide a space for temporarily storing the filtered water produced by the filtration members **118**.

To this end, the support frame **111** may include a loop-shaped edge member **112** having an empty space therein, and a partition member **114** coupled to the edge member **112** so that the empty space may be partitioned into a plurality of storage spaces **115**.

For example, as shown in FIG. 8, the edge member **112** may be a rectangular frame structure with both surfaces open, and the partition member **114** may be a straight member connecting two opposite sides of the edge member **112** to each other.

Further, the partition member **114** may include at least one lateral member **114a** having a predetermined length and at least one longitudinal member **114b** disposed to intersect the lateral member **114a**.

Accordingly, as shown in FIGS. 9 and 10, when the pair of filtration members **118** are attached to both surfaces of the edge member **112**, respectively, the pair of filtration members **118** may be respectively supported by both surfaces of the edge member **112**, and may be maintained in a state spaced apart from each other through the thickness of the edge member **112**.

In this case, the filtration member **118** may be provided in a form of a plate-shaped sheet, and the edge side thereof may be attached to one surface of the edge member **112**.

Accordingly, the filtration member **118** may be maintained in a plate-like shape by the support frame **111**, and may increase the output or flow rate of filtered water by increasing the contact area with the raw water.

However, the shape of the frame member **112** is not limited thereto, and may be appropriately changed to any one of a circular shape, an arc shape, a polygonal shape, and a combination thereof depending on the shape of the filtration member **118**.

In addition, the partition member **114** may be composed of only any one of the lateral member **114a** and the longitudinal member **114b**, and in case any one of the lateral member **114a** and the longitudinal member **114b** is provided in plurality, the plurality of lateral member **114a** or the

plurality of longitudinal member **114b** may be arranged to be spaced apart and parallel to each other.

In this case, the partition member **114** may serve to partition the empty space into a plurality of storage spaces **115** and also support the filtration member **118**.

Accordingly, the edge of the filtration member **118** may be supported by the edge member **112**, and the remaining inner region of the filtration member **118** except for the edge thereof may be supported by the partition member **114**.

As a result, when the filtration member **118** is provided as a plate-shaped sheet having a predetermined area, even if water pressure is applied to the filtration member **118** while the raw water passes through the filtration member **118**, the filtration member **118** may be maintained in a plate-like shape without being bent or deformed by the water pressure.

In this case, the edge member **112** may include a stepped portion **113** formed to be recessed inward to a certain depth along the inner side of the edge, and the stepped portion **113** may be formed on each of both surfaces of the edge member **112**.

In this case, as shown in FIG. 8, a thickness **t2** of the edge member **112** in the portion where the stepped portion **113** is formed may have the same thickness as a thickness **t1** of the partition member **114**.

Accordingly, when the pair of filtration member **118** are respectively attached to both surfaces of the support frame **111**, the edge of the filtration member **118** may be in close contact with the stepped portion **113** as shown in FIG. 9; the inner region of the filtration member **118** may be supported by the partition member **114**; and the edge portion and inner portion of the filtration member **118** may form a horizontal plane.

The at least one communication path **116** may be formed in the partition member **114**.

As shown in FIG. 11, the communication path **116** may connect a plurality of storage spaces **115** formed inside the edge member **112** to each other through the partition member **114**.

Accordingly, the filtered water produced while passing through the filtration member **118** may move toward the receiving port **117** via the plurality of storage spaces **115**.

For example, in case the partition member **114** includes both the lateral member **114a** and the longitudinal member **114b**, the communication path **116** may be formed in each of the lateral member **114a** and the longitudinal member **114b**.

In addition, the communication path **116** may be a through-hole formed through the lateral member **114a** and the longitudinal member **114b**, but may also be an accommodation groove formed to be recessed inwardly from one surface of the lateral member **114a** and the longitudinal member **114b**. In this case, the bottom surface of the communication path **116** may face one surface of the filtration member **118**.

That is, when the pair of filtration member **118** are respectively attached to both surfaces of the support frame **111**, the communication path **116** may be formed on each of both surfaces of the partition member **114** so that the bottom surface thereof may face one surface of the pair of filtration members **118**, respectively.

Accordingly, the filtration member **118** may produce the filtered water not only in an area corresponding to the storage space **115**, but also in a portion of the partition member **114** where the communication path **116** is formed.

Accordingly, even if the partition member **114** supports the inner region of the filtration member **118**, an area in which the filtered water cannot be produced may be minimized, thereby further increasing the output of filtered water.

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Although the drawings show that the communication paths **116** formed on each of both surfaces of the partition member **114** are formed at positions overlapping each other, the formation position of the communication path **116** is not limited thereto, and the position of the communication path formed on one surface of the partition member **114** may not overlap with the position of the communication path formed on the opposite surface.

The filtration member **118** may be provided in a pair as described above and may be attached to each of both surfaces of the support frame **111**.

The above filtration member **118** may filter out foreign substances contained in the raw water while the raw water to be filtered moves from the outside of the filtration member **118** to the storage space **115** by gravity or water pressure.

In this case, the filtration member **118** may be provided as a plate-shaped sheet so as to increase the contact area with the raw water.

Herein, although any known filtration member used for water treatment may be applied as the filtration member **118**, the filtration member **118** may also be a plate-shaped sheet comprising a support **118a** and a nanofiber web layer **119b** disposed on one or both surfaces of the support **118a**.

In the present invention, the nanofiber web layer **119b** may filter out foreign substances contained in the raw water, and the support **118a** may support the nanofiber web layer **119b** and serve as a moving passage for diffusing the filtered water produced by the nanofiber web layer **119b** over the entire area of the filtration member **118**.

As a non-limiting example, the filtration member **118** may have a two-layer structure in which the nanofiber web layer **119b** is directly attached to one surface of the support **118a** as shown in FIG. **15(a)**.

Alternatively, the filtration member **118** may have a three-layer structure in which a pair of nanofiber web layers **119b** are directly attached to both surfaces of the support **118a**, respectively, as shown in FIG. **15(b)**. In this case, even if the nanofiber web layer **119b** attached to the outside of the support **118a** among the nanofiber web layers **119b** attached to both sides of the support **118a** is damaged when washing to remove foreign substances attached to the surface of the filtration member **118**, the nanofiber web layer **119b** attached to the inside of the support **118a** may be maintained in its original shape. Accordingly, the nanofiber web layer **119b** may prevent the filtration efficiency of the filtration member **118** from being lowered, and further enhance the quality of the filtered water by filtering the filtered water passing through the support **118a** again.

In this case, the nanofiber web layer **119b** may be laminated to one surface of the support **118a** through thermal welding, ultrasonic welding, high-frequency welding, etc., and one surface of the filtration member **118** may be attached to one surface of the support frame **111** through ultrasonic welding.

In addition, the support **118a** may be a porous substrate so that the filtered water produced through the nanofiber web layer **119b** may be diffused. For example, the support **118a** may be any one of commonly used known woven, knitted, or non-woven fabrics.

Further, the nanofiber web layer **119b** may be formed of nanofibers to filter out foreign substances contained in the liquid to be filtered, and the nanofiber web layer **119b** may include a nanofiber web of a three-dimensional network structure.

In this case, the nanofiber may include a fiber-forming component, including polyacrylonitrile (PAN) and polyvi-

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nylidene fluoride (PVDF), and an emulsifier for improving the miscibility of the fiber-forming component.

The nanofiber web layer **119b** may be provided as a single layer or as a multilayer.

The receiving port **117** may discharge the filtered water produced through the filtration member **118** from the plurality of storage spaces **115** to the outside.

To this end, the receiving port **117** may be provided on one side of the support frame **111** to communicate with any one of the plurality of storage spaces **115**.

Accordingly, the filtered water produced through the filtration member **118** may move from each storage space **115** to the storage space **115** in communication with the receiving port **117** through the communication path **116** and then be discharged to the outside through the receiving port **117**.

The filtered water integration member **120** may fix one side of each of the plate-shaped filter members **110** so that the plurality of plate-shaped filter members **110** may be maintained in a state spaced apart in parallel with each other at a distance along one direction while facing each other. Further, the filtered water integration member **120** may integrate the filtered water individually produced from each of the plurality of plate-shaped filter members **110**.

That is, the filtered water integration member **120** may serve as a coupling member for fixing each of the plurality of plate-shaped filter members **110**, as a spacer member for maintaining a spaced distance between two plate-shaped filter members **110** spaced apart in parallel with each other, and further as a collecting member for integrating the filtered water produced from each of the plate-shaped filter members **110** into one.

To this end, the filtered water integration member **120** may include a body **121**, a plurality of protrusions **123**, a fastening member **125**, a plurality of inlets **126** and at least one outlet **127**, as shown in FIGS. **12** to **14**, and the plurality of inlets **126** may be connected one-to-one with the receiving ports **117** provided in each of the plate-shaped filter members **110** via a tube **130**.

Specifically, the body **121** may include an integration space or interior manifold volume **122** formed therein to integrate the filtered water incoming from each of the plate-shaped filter members **110** through the inlet **126** connected to the tube **130**, and a plurality of inlets **126** connected to the tube **130** may be formed on one side of the body **121**. In addition, at least one outlet **127** for discharging the filtered water integrated in the integration space **122** to the outside may be formed on one side of the body **121**.

Accordingly, the filtered water produced individually from each of the plate-shaped filter members **110** may be moved to the integration space **122** through the receiving port **117**, the tube **130** and the inlet **126**, and the filtered water introduced into the integration space **122** may be discharged to the outside through the outlet **127**.

In this case, the plurality of protrusions **123** may be formed to protrude at a certain height from one surface of the body **121**, and may be disposed to be spaced apart from each other at a predetermined interval.

Accordingly, an accommodation groove **124** into which one side of the plate-shaped filter member **110** is inserted may be formed between two adjacent protrusions **123**.

Herein, the plate-shaped filter member **110** may include a coupler **111a** formed on one side of the support frame **111** to protrude outward by a certain length, and the coupler **111a** may be inserted into the accommodation groove **124**.

Accordingly, when the coupler **111a** provided in each of the plurality of plate-shaped filter members **110** are respec-

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tively inserted into the plurality of accommodation grooves 124 formed on one side of the body 121, the plurality of plate-shaped filter members 110 may be arranged along one direction so that each one surface thereof faces each other, and be maintained in a state spaced apart from each other by the thickness of the protrusions 123.

As a result, both surfaces of each of the plate-shaped filter members 110 may be in smooth contact with the raw water.

As described above, the filtered water integration member 120 may perform all of a function of integrating the filtered water produced individually through each of the plate-shaped filter members 110 into one, a function of maintaining a spaced distance between two plate-shaped filter members 110 spaced apart in parallel with each other, and a function of fixing each of the plurality of plate-shaped filter members 110.

Herein, the filtered water integration member 120 may include at least one fastening member 125 that is detachably coupled to the body 121, and the at least one fastening member 125 may prevent the plate-shaped filter members 110 respectively inserted into the plurality of accommodation grooves 124 from being separated from the accommodation grooves 124.

For example, the fastening member 125 may be a fastening bar having a predetermined length, and the fastening bar may be fastened to the body 121 to pass through all of the plurality of protrusions 123.

In this case, a fastening hole 128 may be formed through the plurality of protrusions 123 to allow the fastening member 125 to pass therethrough, and a through-hole 119 may be formed through the coupler 111a at a position corresponding to the fastening hole 128.

Accordingly, when the fastening bar is fastened to the body 121 to pass through the fastening hole 128 provided in each of the plate-shaped filter member 110 and the through-hole 119 formed in each of the plurality of protrusions 123 in a state in which the coupler 111a provided in each of the plate-shaped filter members 110 is inserted into the accommodation groove 124, each of the plate-shaped filter members 110 may be fixed to the body 121 in a state in which the coupler 111a is inserted into the accommodation groove 124.

Through this, the plate-shaped filter members 110 respectively inserted into the plurality of accommodation grooves 124 may be prevented from being separated from the accommodation groove 124 by the fastening bar.

Accordingly, the total number of use of the plate-shaped filter member 110 fixed to the body 121 in the filter module 100 may be easily adjusted according to the production of filtered water required per hour.

In addition, each of the plate-shaped filter members 110 in the filter module 100 may be detachably coupled to the body 121, whereby only the plate-shaped filter members requiring replacement among the plurality of plate-shaped filter members 110 may be individually replaced.

Accordingly, the gravity-type water-purifying device 1000 according to an embodiment of the present invention may improve convenience of use and reduce maintenance costs.

That is, since the plurality of plate-shaped filter members 110 and the filtered water integration member 120 in the filter module 100 may be configured as individual modules that may be detached from each other, some destroyed or damaged parts may be individually replaced. Accordingly, the gravity-type water-purifying device 1000 according to an embodiment of the present invention may reduce maintenance costs.

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Although an embodiment of the present invention have been described above, the spirit of the present invention is not limited to the embodiment presented in the subject specification; and those skilled in the art who understands the spirit of the present invention will be able to easily suggest other embodiments through addition, changes, elimination, and the like of elements without departing from the scope of the same spirit, and such other embodiments will also fall within the scope of the present invention.

The invention claimed is:

1. A gravity-flow water purifying device comprising:
 - a housing (200) defining an interior volume (210), the housing (200) having a housing outlet (220) at a bottom of the housing (200);
 - a filter module (100) disposed within the interior volume (210), including:
 - a plurality of filter plates (110), each plate defining an interior empty space for receiving filtrate;
 - a manifold (120) defining an interior manifold volume (122), the manifold (120) having a plurality of inlets (126) in fluid communication with the interior manifold volume (122), wherein each inlet (126) of the plurality of inlets (126) is in fluid communication with the interior empty space of a respective filter plate (110), the manifold (120) having at least one outlet (127) in fluid communication with the housing outlet (220), a top of the manifold (120) having a plurality of parallel accommodation grooves (124) each sized and shaped to receive and support a filter plate (110); and
 - a movable storage vessel (1210) disposed below the housing (200), the movable storage vessel (1210) having an inlet opening in fluid communication with the at least one outlet (127) for receiving filtered water.
2. The gravity-flow water purifying device of claim 1, wherein the movable storage vessel (1210) is box-shaped and includes a fitting (1220) formed in the movable storage vessel so as to be detachably coupled to an end of a connection pipe (1300).
3. The gravity-flow water purifying device of claim 1, wherein an additional filter (1400) is disposed in a connection pipe (1300) fluidically disposed between the at least one outlet (127) and the inlet of the movable storage vessel for further filtering water exiting the housing (200).
4. The gravity-flow water purifying device of claim 3, wherein the connection pipe (1300) includes a valve (1500) for controlling flow therethrough.
5. The gravity-flow water purifying device of claim 1, wherein each of the plurality of filter plates (110) includes:
 - a support frame (111) comprising a loop-shaped edge member (112) defining the empty space which is open on both sides, and a partition member (114) coupled to the loop-shaped edge member (112) to partition the empty space into a plurality of storage spaces (115);
 - a pair of filtration members (118) shaped to conform with the shape of the support frame (111), having a predetermined area and being coupled to both surfaces of the support frame (111) to produce filtered water or filtrate from unfiltered water moving from the outside of each filter plate (110) to the plurality of storage spaces (115);
 - at least one communication path (116) formed in the partition member (114) to allow the plurality of storage spaces (115) to be in fluid communication with each other; and
 - a receiving port (117) provided on one side of the support frame (111) and in fluid communication with the plurality of storage spaces (115) and in fluid communication

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tion with the interior manifold volume (122) of the manifold (120) via a tube (130) configured to deliver filtered water from the storage spaces (115) to the interior manifold volume (122).

6. The gravity-flow water purifying device of claim 5, wherein the communication path (116) is at least one recessed portion in the partition member (114).

7. The gravity-flow water purifying device of claim 5, wherein the communication path (116) is at least one through-hole in the partition member (114).

8. The gravity-flow water purifying device of claim 5, wherein the edge member (112) includes a stepped portion (113) formed along the inner side of the edge, and an edge side of the filtration member (118) is fixed to the stepped portion (113).

9. The gravity-flow water purifying device of claim 5, wherein the partition member (114) includes at least one lateral member (114a) having a predetermined length and at least one longitudinal member (114b) disposed to intersect the lateral member.

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10. The gravity-flow water purifying device of claim 5, wherein the filtration member (118) includes a plate-shaped support (118a) having a predetermined area, and a nanofiber web layer (119b) formed of nanofiber and disposed on one surface or both surfaces of the plate-shaped support (118a).

11. The gravity-flow water purifying device of claim 5, wherein all of the plurality of filter plates (110) comprising filter members made of the same filtration material.

12. The gravity-flow water purifying device of claim 1, further comprising:

at least one fastening member (125) removably extending through the structure defining the plurality of parallel accommodation grooves (124) and through a coupler (111a) protruding from each filter plate frame (111), to couple the filter plates (110) to the manifold (120), wherein each coupler (111a) extends into a respective parallel accommodation groove (124).

13. The gravity-flow water purifying device of claim 12, wherein the at least one fastening member (125) is a bar or bolt.

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